



Deep Learning Final Project

Federated Image Classification

under Non-IID Data

Group 9 - Topic 27

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Abbreviations

CNN: convolutional neural network. FL: federated learning. IID: independent and identically distributed. SGD: stochastic gradient descent. CE: cross entropy. TV: total variation. SD: sample standard deviation. E: local epochs. R: communication rounds. pp: percentage points.

Abstract

This project investigates how client label distributions affect federated image classification. A small convolutional neural network and a manually implemented FedAvg algorithm are evaluated on Fashion-MNIST with ten simulated clients. An exact quota design produces IID, mild and strong label skew while keeping client sample counts and global class totals fixed. Five configurations are evaluated over three seeds, with 1,620,000 training-image exposures per run. Final test accuracy is 90.13% for centralized training, 83.25% for IID FedAvg, 81.93% for mild skew and 77.73% for strong skew. Under strong skew, three local epochs and ten rounds achieve 77.77% accuracy, close to one local epoch and thirty rounds at the same exposure budget. Error analysis identifies Shirt as the weakest class and examines traceable Shirt-to-T-shirt/top mistakes. Saved predictions, split indices and experiment manifests support an independent integrity audit. The findings demonstrate sensitivity to label skew within this controlled setting, while the local-epoch comparison suggests a potential aggregation trade-off. Conclusions remain limited to one dataset, one CNN, full participation and three seeds. The simulation supplies no formal privacy guarantee.

Keywords: federated learning; image classification; non-IID data; FedAvg; Fashion-MNIST.

I. Introduction and Research Question

1. Problem statement

Images collected by different organizations can have very different label distributions. A clothing retailer specializing in footwear, for example, may hold few examples of shirts. Centralized training would combine these images into one dataset. Federated learning instead trains a shared classifier through local model updates, while each client retains its training examples [1]. The resulting optimization problem depends on what each client observes between aggregation steps.

This project studies **label skew**: clients have equal sample counts but different proportions of the ten Fashion-MNIST classes. Equal quantities make distribution differences easier to interpret. A centralized CNN provides a reference, and FedAvg provides the main method. All clients participate in every round in a single-machine simulation.

2. Research questions and contributions

RQ1: How does FedAvg with IID client data compare with centralized training under the same image-exposure budget? **RQ2:** How do mild and strong label skew change final accuracy, macro-F1 and learning curves? **RQ3:** Under strong skew, can three local epochs and fewer aggregation rounds retain performance at fixed training exposure?

The project delivers a transparent PyTorch implementation, an exact partition design, fifteen recorded runs and an offline inference demo. Its contribution is a controlled comparison and traceable error analysis. It does not introduce a new optimization algorithm. The experiments support conclusions within this dataset, CNN and participation protocol.

II. Related Work

1. Federated averaging

McMahan et al. [1] introduced federated averaging, in which clients train local copies of a common model and the server combines their updated parameters according to sample counts. Their experiments establish model averaging as a practical approach to decentralized learning. Our implementation follows this central mechanism. It uses ten simulated clients and full participation so that the effect of label distributions can be examined without varying client sampling.

An important distinction is that one local epoch contains many minibatch updates. Averaging the resulting parameter vectors does not reproduce SGD on globally shuffled minibatches. Consequently, even the IID setting may differ from centralized learning. We investigate that difference empirically rather than assuming that IID FedAvg must match its centralized reference.

2. Statistical heterogeneity

Hsu, Qi and Brown [3] study the effects of non-identical distributions in federated visual classification. Their work motivates treating heterogeneity as a controlled experimental variable rather than a binary property. Here, an explicit quota matrix provides three known severity levels while retaining equal numbers of images per client and equal global class totals.

FedProx [4] adds a proximal term to local objectives to address heterogeneity. SCAFFOLD [5] uses control variates to correct client drift. These approaches show why divergence between local optimization trajectories matters and suggest useful future comparisons. Neither method is implemented or evaluated in this project. We use FedAvg as the only federated optimizer so that the experiment remains focused on the data distribution.

3. Dataset and privacy context

Xiao, Rasul and Vollgraf [2] provide Fashion-MNIST, a ten-class benchmark with small grayscale images. It makes repeated controlled experiments feasible while retaining visually overlapping categories. Clothing categories such as Shirt and T-shirt/top also support meaningful qualitative error analysis.

Zhu, Liu and Han [6] demonstrate that shared gradients can reveal training information under particular conditions. Our server aggregation API receives model tensors and sample counts, but this design alone supplies no formal privacy guarantee. We do not test attacks or implement differential privacy or secure aggregation. The simulation concerns collaborative optimization and classification quality.

4. Position of this study

Unlike a study that changes model size or dataset together with heterogeneity, this comparison fixes the CNN, normalization, optimizer and training pool. The quota design is a project-specific experimental construction. It is not presented as the original partitioning method of any cited paper. Three seeds measure preliminary run-to-run variation, and conclusions remain descriptive.

III. Dataset and Data Preparation

1. Dataset identity and version

Fashion-MNIST [2] contains 60,000 official training images and 10,000 official test images. Each image has one grayscale channel, resolution 28 x 28 and one of ten labels. We use the original 2017 Fashion-MNIST IDX distribution, identified by the four official gzip filenames and their MD5 checksums. There is no invented semantic release number. The authoritative dataset and license are available at <https://github.com/zalandoresearch/fashion-mnist>.

Label	Class	Label	Class
0	T-shirt/top	5	Sandal
1	Trouser	6	Shirt
2	Pullover	7	Sneaker
3	Dress	8	Bag
4	Coat	9	Ankle boot

Table 1. The ten official label meanings, preserved throughout the pipeline.

2. Split and preprocessing

A fixed stratified split with seed 2026 reserves 600 images per class for validation. The remaining 5,400 per class form the training pool. Thus training, validation and test contain 54,000, 6,000 and 10,000 images, respectively. The official test split remains independent and contains 1,000 examples of each class. Persisted indices use the order of the original official files.

Images become float tensors with shape $N \times 1 \times 28 \times 28$ and normalization $(\text{uint8} / 255 - 0.5) / 0.5$. The resulting range is $[-1, 1]$. There is no resizing, augmentation or fitted normalization statistic. The same transform applies to centralized and federated learning. No validation or test image enters a training minibatch.

3. Reproducing the processed data

`experiments/prepare_data.py` downloads the official files over HTTPS, verifies their official MD5 checksums, reproduces the split and saves all client partitions without training. `DATA.md` documents the dataset version, URLs, checksums, commands, label mapping and licensing. Processed tensors are deterministic transformations of official images, not a separately curated dataset.

The processed data are reproducible from official files and committed indices. Public release v1.0.0 provides `processed-data.zip` with all 70,000 normalized images, labels, the 54k/6k split and nine client partitions. `DATA.md` links the archive and records its loading procedure; the release manifest verifies SHA-256. A separate 100-image test subset supports the demo. Dataset caches are excluded from Git.

IV. Controlled Client Partitioning

1. Balanced cyclic label skew

For each run seed, a deterministic class permutation defines the dominant-label set S_k : the two consecutive labels assigned to client k , with cyclic wraparound. Each label is dominant at exactly two clients. For $C = 10$ classes, the prescribed local class proportion is:

$$p_k(c) = \frac{1-\lambda}{C} + \frac{\lambda}{2} \mathbf{1}_{\{c \in S_k\}}, \quad C = 10. \quad (1)$$

Here $p_k(c)$ is the proportion of class c at client k . The indicator is one for a dominant class and zero otherwise. The skew parameter λ is 0, 0.5 or 0.9. Each client has 5,400 images; its class quota is $5,400 p_k(c)$. The construction preserves a uniform global class prior.

Setting	Lambda	Dominant class	Other class	Images/client	TV
IID	0.0	540	540	5,400	0.00
Mild	0.5	1,620	270	5,400	0.40
Strong	0.9	2,484	54	5,400	0.72

Table 2. Exact training quotas; each client has two dominant and eight other classes.

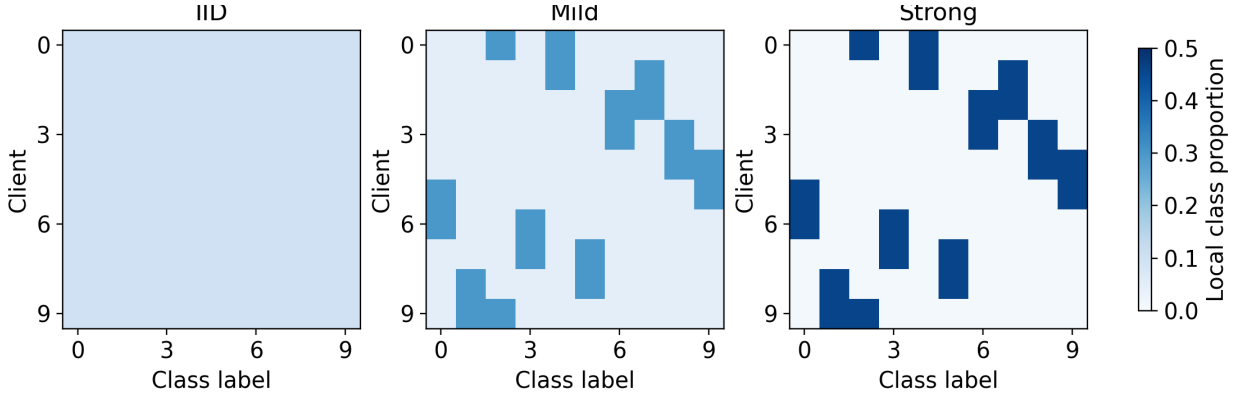


Figure 1. Client class-proportion heatmaps for the three severities, illustrated with seed 42.

2. Integrity and interpretation

Each matrix row and column sums to 5,400. Shuffled indices are allocated without replacement, so disjoint client sets exactly cover the training pool. The class permutation is fixed across severities within a seed; both class pairings and image assignments vary between seeds.

Total variation measures the distance between a client's label proportions and the uniform class distribution:

$$\text{TV}_k = \frac{1}{2} \sum_{c=0}^{C-1} \left| p_k(c) - \frac{1}{C} \right| = 0.8\lambda. \quad (2)$$

Eq. (2) sums two dominant-class deviations of 0.4λ and eight others of 0.1λ . Thus TV is 0, 0.40 or 0.72. Strong clients retain all ten classes, with 92% of images in two dominant labels. This synthetic study excludes camera-domain shifts, label corruption and unequal quantities.

V. Methods

1. Centralized baseline and shared CNN

The centralized CNN trains on all 54,000 images with the same preprocessing, architecture, seed, loss, learning rate and thirty passes as the federated runs. It provides a protocol-specific reference.

Layer	Configuration	Output shape	Parameters
Input	Grayscale image	1 x 28 x 28	0
Conv1 + ReLU	1 to 16, 3 x 3, padding 1	16 x 28 x 28	160
MaxPool	2 x 2, stride 2	16 x 14 x 14	0
Conv2 + ReLU	16 to 32, 3 x 3, padding 1	32 x 14 x 14	4,640
MaxPool	2 x 2, stride 2	32 x 7 x 7	0
Flatten	32 x 7 x 7	1,568	0
Linear + ReLU	1,568 to 64	64	100,416
Linear	64 to 10	10 logits	650
Total	SmallCNN	10 logits	105,866

Table 3. Architecture and parameter count, including biases.

Convolutions, ReLU and pooling produce features for ten class scores. Training starts from scratch, with no pretrained weights, BatchNorm or dropout adding state to aggregation.

2. Objective and optimizer

For the ten logits z produced by the CNN and correct label y , the per-image cross-entropy loss is:

$$\ell(z, y) = -z_y + \log \left(\sum_{c=0}^{C-1} e^{z_c} \right) = -\log p_w(y | x). \quad (3)$$

Here w denotes CNN parameters, x the image and p_w softmax probabilities. CrossEntropyLoss averages Eq. (3) over raw logits in a minibatch; inference selects the largest logit.

Both training methods use the same SGD update. For client k , round t and local minibatch step s :

$$w_{t,s+1}^{(k)} = w_{t,s}^{(k)} - \eta \nabla_w \left[\frac{1}{|\mathcal{B}_{k,t,s}|} \sum_{(x,y) \in \mathcal{B}_{k,t,s}} \ell(f_w(x), y) \right]_{w=w_{t,s}^{(k)}}. \quad (4)$$

Minibatches contain up to 64 images; $\eta = 0.01$. Index s counts minibatch steps; E counts full local epochs. Momentum and weight decay are zero. Partial batches are retained, and local weights reset to the global snapshot each round.

3. Comparison strategy

Final global models share equal initialization hashes across all five settings within each seed. Pairing controls initialization, while data ordering and optimization trajectories may differ. Accuracy, macro metrics and cross entropy describe complementary aspects of classification quality.

V. Methods (continued)

4. Main method: FedAvg

Let D_k contain n_k training examples at client k . With $K = 10$ clients indexed from 0 to $K - 1$, local and global empirical losses are:

$$F_k(w) = \frac{1}{n_k} \sum_{(x,y) \in D_k} \ell(f_w(x), y), \quad F(w) = \sum_{k=0}^{K-1} \frac{n_k}{N} F_k(w). \quad (5)$$

Here $f_w(x)$ is the CNN logit vector and $N = 54,000$ is the total training size. Every client starts from the same global snapshot w_t and performs E local epochs using Eq. (4).

The server aggregates these client models using FedAvg [1]:

$$w_{t+1} = \sum_{k=0}^{K-1} \frac{n_k}{N} w_{t+1}^{(k)} = \frac{1}{10} \sum_{k=0}^9 w_{t+1}^{(k)}. \quad (6)$$

The returned local weights in Eq. (6) receive weight 0.1 because every client participates with $n_k = 5,400$. General sample-count weighting is tested with unequal counts. Aggregation averages all floating parameter tensors, including biases, to produce one global model.

5. Round procedure

```
initialize the global SmallCNN with the run seed
for round t = 1, ..., R:
    snapshot = clone(global_model.state_dict())
    local_states = []
    for client k = 0, ..., 9:
        local_model.load_state_dict(snapshot)
        train local_model on D(k) for E epochs
        append cloned local state and n(k)
    global_state = weighted_average(local_states, sample_counts)
    load global_state and evaluate fixed validation data
    save the complete round checkpoint and history
```

Listing 1. FedAvg control flow; test evaluation occurs after training.

Snapshots and returned states are independent copies, preventing aliasing during aggregation. Sequential client execution is scheduling only: each client receives the same global starting point, rather than the previous client's trained weights.

6. State, evaluation and resume

`server.py` coordinates clients, `client.py` resets local state, and `fedavg.py` aggregates tensors and sample counts, checking keys, shapes, dtypes and finite values. The simulation loads all data to construct subsets; this API does not establish physical isolation across machines.

Checkpoints retain global weights, configuration and history. Data-order seeds depend on the run seed, round, client and epoch. Resume restarts from the last completed round and discards partial-round work; a test verifies the final parameter hash against uninterrupted training. The demo performs inference from a trained global checkpoint.

VI. Experimental Setup

1. Three required setups

Setup 1: baseline versus main method (RQ1). Compare centralized CNN with FedAvg IID on the same 54,000 images and thirty passes. Their minibatch mixing and update coordination differ.

Setup 2: main research experiment. Compare FedAvg IID, mild and strong label skew with $E=1$ and $R=30$. This changes the client distribution while holding model, optimizer, quantities, participation and budget fixed. It answers RQ2.

Setup 3: ablation and seed robustness (RQ3). Compare strong E1R30 and E3R10, sharing partition and initialization within a seed. E and R change together at fixed exposure. Five settings and seeds 42, 43 and 44 produce fifteen completed runs.

Configuration	Lambda	Local epochs E	Rounds / epochs	Exposures/run
Centralized	-	-	30 epochs	1,620,000
FedAvg IID	0.0	1	30 rounds	1,620,000
FedAvg mild	0.5	1	30 rounds	1,620,000
FedAvg strong	0.9	1	30 rounds	1,620,000
Strong E3R10	0.9	3	10 rounds	1,620,000

Table 4. Locked experiment matrix. Each row uses all three run seeds.

2. Fairness and budget accounting

With full participation, R rounds and E local epochs per round, the number of training-image exposures is:

$$B = RE \sum_{k=0}^{K-1} n_k = REN = 30 \times 54,000 = 1,620,000. \quad (7)$$

Both E1R30 and E3R10 have $RE = 30$; centralized training also makes 30 passes. Across fifteen runs the total is 24,300,000 exposures. Retaining partial batches gives 25,320 centralized and 25,500 federated optimizer steps per run. Equal image exposure therefore allows a slight difference in optimizer-step counts.

Equal exposure does not imply identical optimization or runtime: aggregation frequency and minibatch composition differ. The E3 comparison changes local epochs and rounds together, measuring aggregation frequency at fixed exposure.

3. Execution environment

The recorded versions are PyTorch 2.8.0, torchvision 0.23.0, NumPy 2.2.6 and scikit-learn 1.7.2. The reproduction instructions target Python 3.12.

Runs use one CPU thread per process on macOS arm64. Concurrent processes may affect wall time; no hardware-speed or network-latency claim is made. Manifests retain the training commit, library versions, initial hash, configuration and partition identity.

VI. Experimental Setup (continued)

4. Evaluation metrics

Let M be the confusion matrix, with true labels in rows and predicted labels in columns, and $N_{\text{test}} = 10,000$. Test accuracy is:

$$\text{Accuracy} = \frac{1}{N_{\text{test}}} \sum_{c=0}^{C-1} M_{cc}. \quad (8)$$

For class c , $TP_c = M_{cc}$, FP_c counts false positives and FN_c counts false negatives. Precision, recall and macro-F1 are:

$$P_c = \frac{TP_c}{TP_c + FP_c}, \quad R_c = \frac{TP_c}{TP_c + FN_c}. \quad (9)$$

$$\text{MacroF1} = \frac{1}{C} \sum_{c=0}^{C-1} F1_c, \quad F1_c = \frac{2P_c R_c}{P_c + R_c}, \quad C = 10. \quad (10)$$

Zero denominators give zero scores. Macro precision and recall average class scores; equal test support makes macro recall equal accuracy. Test loss averages Eq. (3) over all images.

5. Checkpoint and test protocol

Primary results use the final-budget checkpoint. Secondary results use highest validation accuracy, with earliest-point ties. Testing follows training and validation selection; test curves are descriptive and do not select hyperparameters or checkpoints.

Auditing the recorded outputs verifies consistency and coverage, without establishing every historical decision about inspecting test results. No new hyperparameter selection was performed.

6. Variability and convergence indicator

Across seeds 42, 43 and 44, the metric mean and sample standard deviation are:

$$\bar{x} = \frac{1}{3} \sum_{i=1}^3 x_i, \quad s = \sqrt{\frac{1}{3-1} \sum_{i=1}^3 (x_i - \bar{x})^2}. \quad (11)$$

Here x_i is run i 's metric. Tables report mean $\pm$ s, with accuracy in percent and its differences in percentage points. Other scales are unchanged; Appendix B lists all runs.

R@80 starts the first three consecutive post-initialization validation points at or above 80%, or is not reached. It is descriptive, not a convergence proof. Exposure differs for E1 and E3, requiring epoch context.

7. Integrity and reproducibility checks

The audit checks all 10,000 predictions per run, budgets, paired initialization, partitions, split coverage and histories. Tests cover aggregation, aliasing, snapshot resets, metrics, CNN shape and size, tiny-dataset learning and deterministic resume.

VII. Results and Discussion

1. Final-budget comparison

Setting	Accuracy (%)	Macro precision	Macro-F1	Test CE
Centralized	90.13 +/- 0.14	0.9013 +/- 0.0007	0.9004 +/- 0.0013	0.2756 +/- 0.0072
FedAvg IID	83.25 +/- 0.58	0.8357 +/- 0.0035	0.8317 +/- 0.0054	0.4588 +/- 0.0106
FedAvg mild	81.93 +/- 0.18	0.8169 +/- 0.0025	0.8154 +/- 0.0026	0.4875 +/- 0.0039
FedAvg strong	77.73 +/- 0.92	0.7690 +/- 0.0167	0.7685 +/- 0.0172	0.5839 +/- 0.0176
Strong E3R10	77.77 +/- 0.60	0.7712 +/- 0.0084	0.7711 +/- 0.0099	0.5838 +/- 0.0196

Table 5. Official-test metrics, mean +/- sample SD across seeds 42, 43 and 44. Macro recall equals accuracy expressed on $[0, 1]$. Full per-seed metrics are retained in CSV.

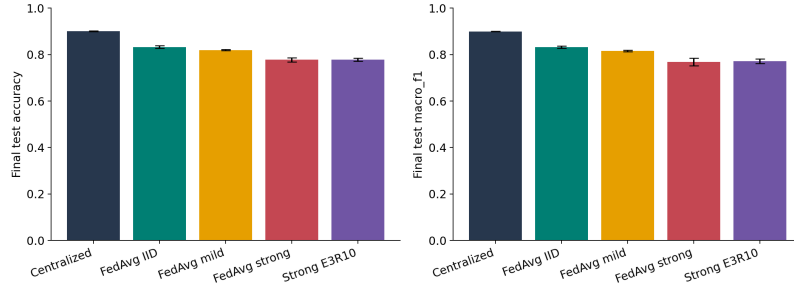


Figure 2. Final-budget accuracy and macro-F1 with sample-SD error bars across three seeds.

2. Baseline versus federated training: RQ1

Centralized accuracy is 90.13% compared with 83.25% for IID FedAvg, a gap of 6.88 percentage points. Macro-F1 is 0.9004 and 0.8317, respectively. Equal exposure therefore does not remove the difference between the two recorded optimization procedures.

Centralized minibatches can combine examples from the complete training pool at each update. FedAvg combines ten parameters after each client has performed many steps, and nonlinear local trajectories generally do not average into the trajectory of centralized SGD. This provides a plausible explanation for the observed gap. The experiment does not identify a unique cause or establish that centralized learning is always superior under every FL protocol.

3. Severity comparison: RQ2

Mild skew reduces mean accuracy by 1.33 percentage points relative to IID. Strong skew reduces it by 5.52 points. The corresponding macro-F1 values are 0.8154 and 0.7685. The means decrease with severity across these three settings, but this finite comparison does not establish a universal monotonic relationship.

Both accuracy and macro-F1 worsen under strong skew, and cross entropy increases. Their agreement suggests the effect is not confined to a single aggregate score. Within this construction, equal sample counts and global label totals make local label proportions the intended explanatory variable. Three seeds give preliminary evidence about repeatability, without a formal population-level significance claim.

VII. Results and Discussion (continued)

4. Learning progress and stability

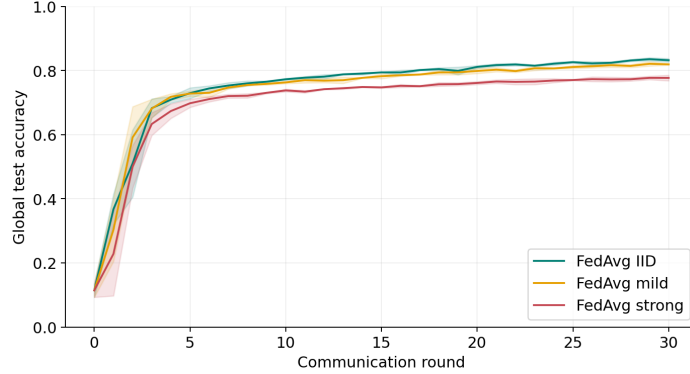


Figure 3. Post-training test accuracy for IID, mild and strong FedAvg. The horizontal axis counts rounds. Shading shows sample SD.

The curves retain the initial model evaluation at point zero. They are obtained from checkpoints after training. They describe the completed trajectories and are not used as an oracle for model selection. Interpreting them alongside the final table avoids reducing the study to one endpoint.

Setting	Seeds reaching	Mean point	Effective epochs
Centralized	3 / 3	2.00	2.00
FedAvg IID	3 / 3	15.33	15.33
FedAvg mild	3 / 3	19.67	19.67
FedAvg strong	0 / 3	Not reached	Not reached
Strong E3R10	0 / 3	Not reached	Not reached

Table 6. First qualifying validation milestone, averaged only over seeds that reached it. Centralized points are epochs; federated points are rounds.

The IID runs reach the 80% validation criterion earlier than mild runs. Neither strong configuration reaches it under the recorded definition. This is consistent with harder global optimization under skew, although the milestone depends on the selected threshold and finite budget. Failure to reach the milestone does not imply that learning stopped or that eventual convergence is impossible.

5. Distinguishing local and global losses

Weighted local training loss averages the cross entropy encountered while client models change during a round. Global validation loss evaluates one aggregated model on the fixed validation set. They differ in evaluated weights, examples and timing. A decrease in local loss can coexist with weak global performance when clients fit their dominant labels.

The initial log stores zero as a sentinel for local training loss because no training occurred at round zero. The plotting code excludes that sentinel from local-loss curves. Appendix B links the loss figures and raw history columns. Validation curves should be read with accuracy and class-level errors, rather than used alone to attribute all changes to client drift.

VII. Results and Discussion (continued)

6. Local-epoch ablation: RQ3

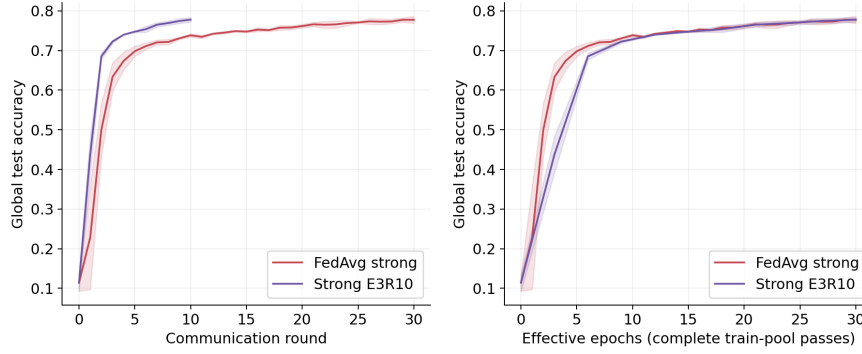


Figure 4. Strong-skew E1R30 and E3R10 under equal image exposure. The effective-epoch view preserves the compute comparison.

The E3R10 configuration reaches 77.77% test accuracy, compared with 77.73% for E1R30. The difference is +0.04 percentage points. Their test cross entropies are 0.5838 and 0.5839, and macro-F1 values are 0.7711 and 0.7685. The mean accuracy difference is much smaller than either setting's seed SD.

The change removes twenty aggregation rounds while maintaining thirty effective passes over the training pool. In a deployed system this could reduce the number of synchronization events. This simulation counts rounds but measures no transmitted bytes, network delay or energy consumption, so it cannot quantify practical communication savings beyond that count.

7. Paired seed differences

Seed	Strong E1R30 (%)	Strong E3R10 (%)	Difference (pp)
42	76.74	77.13	+0.39
43	78.55	78.32	-0.23
44	77.90	77.87	-0.03

Table 7. E3 minus E1 final test accuracy within each seed, in percentage points.

The paired differences are small and not uniformly positive. Accordingly, the result supports similar observed mean performance, not proven equivalence or a reliable gain from E3. With only three seeds, the variation can conceal modest effects.

Longer local training can move parameters further toward each client objective before averaging. Client drift is therefore a plausible trade-off [5]. We did not record gradient cosine similarity, update disagreement or personalized validation checkpoints, so those mechanisms remain hypotheses. Future work should measure update norms and compare FedProx or SCAFFOLD under the same partitions and budget.

The main severity study fixes E and R. The ablation couples them to compare aggregation frequency at fixed exposure; it does not isolate the effect of E alone.

VIII. Error and Qualitative Analysis

1. Class-level weaknesses

For strong E1R30, Shirt has the lowest mean recall, 0.2690 across the three seeds. In seed 42, the largest off-diagonal confusion entry is Shirt predicted as T-shirt/top, with 267 of the 1,000 Shirt test images. This is an observed error pattern, not proof that label skew alone causes all Shirt mistakes.

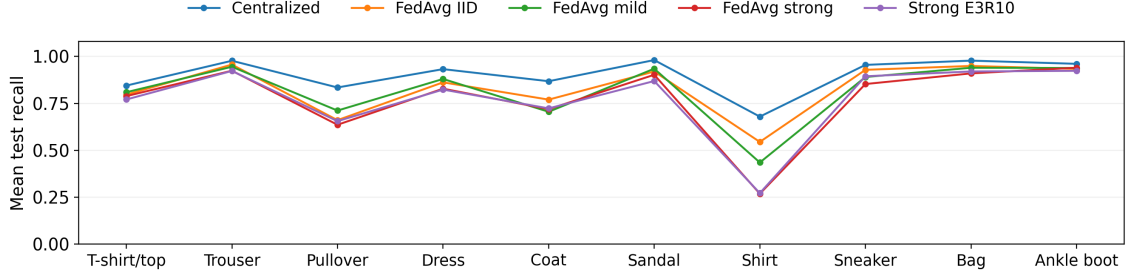


Figure 5. Class recall at the final checkpoint, averaged across three seeds for every setting.

The Shirt class includes upper-body garments that can resemble T-shirt/top, Coat and Pullover in 28 x 28 grayscale images. Ambiguous outlines and loss of texture at this resolution are plausible sources of confusion. Strong client skew may further reduce the regular mixing of competing categories. These explanations are interpretive: label quality and visual ambiguity were not independently annotated, and drift was not directly measured.

2. Actual error examples

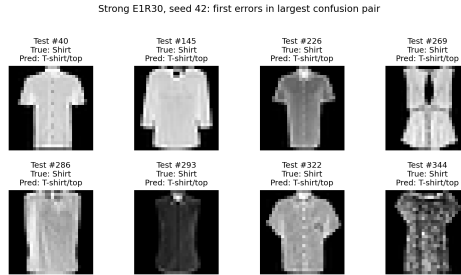


Figure 6. Eight actual Shirt-to-T-shirt/top errors from strong E1R30 seed 42. Images are from Fashion-MNIST, Zalando Research, MIT license.

The examples are the first eight matching errors in official test order, with indices **40, 145, 226, 269, 286, 293, 322 and 344**. Ground truth is label 6 (Shirt) and prediction is label 0 (T-shirt/top) for every image. Their traceable selection prevents arbitrary illustrative images from being substituted for recorded model failures. The confusion matrix and error index table are retained with the experiment outputs.

3. Local validation interpretation

Each client's separate validation subset contains 600 images and follows its training label proportions with disjoint indices. Under strong skew, each dominant class contributes 276 validation images and each other class only six. The same final global model is evaluated on these subsets. These are distribution-specific assessments, not ten personalized models. Client IDs pair different classes across seeds, so averages by client ID require caution.

IX. Conclusion and Limitations

1. Conclusion

Within the fixed protocol, IID FedAvg trails the centralized reference by 6.88 accuracy points. Strong label skew reduces FedAvg accuracy by 5.52 points relative to IID. The E3R10 ablation gives 77.77% mean accuracy versus 77.73% for E1R30 with twenty fewer aggregation rounds at the same exposure budget. These findings answer the three research questions descriptively: optimization coordination matters, stronger local label concentration is associated with poorer global classification, and fewer aggregations can retain similar observed mean performance in this experiment.

The pipeline supplies reproducible data preparation, training, evaluation and inference. All fifteen recorded runs have traceable predictions and manifests.

2. Limitations

One dataset, one small CNN and three seeds limit generalization and statistical strength. Balanced synthetic label skew excludes quantity imbalance, domain shifts and unavailable clients. Full participation and a single-machine simulation do not represent deployed networks or physical data isolation.

Image exposure is matched, while optimizer steps differ slightly. Other learning rates or budgets could change the comparison. Client drift is a plausible explanation but was not measured directly. Shared updates supply no formal privacy guarantee. Concurrent CPU wall times cannot establish network savings or hardware speedups.

3. Future work

Measure client update disagreement, then compare FedProx and SCAFFOLD under the same partitions and exposure budget. Evaluate quantity imbalance and partial participation separately. Secure aggregation and differential privacy require explicit mechanisms and a distinct privacy-versus-utility study.

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Appendix A. Member Contribution Table

The following table records responsibilities for final integration and verification. The group confirmed these final responsibility assignments.

Member	Student ID	Responsibility
Đổng Sỹ Nguyên	23BA14219	FedAvg implementation and final integration review
Đỗ Xuân Nguyên	23BA14217	Dataset provenance, split and preprocessing review
Nguyễn Bình Minh	23BA14197	Non-IID partition and quota validation review
Nguyễn Hồng Quân	23BA14237	CNN architecture and training-loop review
Nguyễn Duy Dũng	23BA14069	Experiment reproduction, submission coordination and group leadership
Đoàn Trung Dũng	23BA14070	Metrics, report and inference-demo verification

Group leader: Nguyễn Duy Dũng (23BA14069).

Appendix B. Per-Seed Results

Setting	Seed 42	Seed 43	Seed 44	Mean +/- SD
Centralized	90.12	90.00	90.28	90.13 +/- 0.14
FedAvg IID	82.63	83.77	83.36	83.25 +/- 0.58
FedAvg mild	81.72	82.01	82.05	81.93 +/- 0.18
FedAvg strong	76.74	78.55	77.90	77.73 +/- 0.92
Strong E3R10	77.13	78.32	77.87	77.77 +/- 0.60

Table B1. Final official-test accuracy (%), unrounded means computed from the stored run metrics. Sample SD uses ddof=1.

Setting	Mean test acc. (%)	SD (%)	Selected points 42/43/44
Centralized	90.03	0.17	30/29/28
FedAvg IID	83.51	0.44	29/30/29
FedAvg mild	82.23	0.61	30/29/30
FedAvg strong	77.79	0.48	29/29/29
Strong E3R10	77.77	0.60	10/10/10

Table B2. Secondary test accuracy at the checkpoint selected by highest validation accuracy, with earliest-point tie-breaking. Selection never uses maximum test accuracy.

Appendix B. Reproduction and Artifact Index (continued)

1. Installation and data preparation

```
git clone https://github.com/nguyends23ba14219-code/DL2026-9-27.git
cd DL2026-9-27
python3.12 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements-dev.txt
python experiments/prepare_data.py --output-dir outputs/reproduction
python -m pytest -q
```

2. Training and evaluation

```
python main.py --config configs/iid.yaml --seed 42 --device cpu \
  --output-dir outputs/reproduction
python experiments/run_suite.py --device cpu \
  --output-dir outputs/reproduction
python experiments/analyze.py --output-dir outputs/reproduction
python experiments/audit.py --output-dir outputs/reproduction
```

The full suite contains all five configurations and three seeds. Resume uses the same configuration and output directory with `--resume`. A new output directory keeps a reproduced study distinct from committed official outputs. Training may take substantially longer on slower hardware.

3. Original study and demo

```
python experiments/audit.py
python experiments/fetch_artifacts.py
streamlit run demo/app.py
# Retrieve every recorded round for checkpoint inspection:
python experiments/fetch_artifacts.py --full
# Rebuild the submission PDF:
python -m pip install -r requirements-docs.txt
python experiments/build_submission.py
```

The demo uses trained global checkpoints and a fixed 100-image official-test subset. It performs inference, not live training. The release manifest records SHA-256 hashes. Full checkpoints cover the initial model and every completed round. The demo remains runnable offline after dependencies and artifacts have been downloaded.

4. Evidence locations

`outputs/splits/seed_2026.npz` stores the fixed split. `outputs/partitions/` stores client indices and count manifests. Each folder in `outputs/runs/` holds configuration, source/environment manifest, training history, post-hoc test history, final metrics and predictions. `outputs/tables/` contains summary, per-seed, class/client and error tables. `outputs/figures/` contains the corresponding plots, including losses and confusion matrices.

`outputs/audit.json` records integrity checks; the release manifest identifies public model/data archives. `DATA.md` documents loading and reproduction. Run `python experiments/fetch_artifacts.py --data` to verify and extract the complete dataset. The four-slide overview lasts three minutes, with twelve minutes of questions. Members should explain the split, round snapshot, weighting and ablation limits.

PyTorch [7] cautions that results vary by platform and release. Seeds and artifacts support reproduction in the recorded environment. CPU, CUDA and MPS are supported; official results use CPU.